

Satoru Akita

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📝 Blog (55+ posts)

SUMMARY

- Software Engineer with **7+ years at Sony Semiconductor**, focused on edge AI, cloud-connected AI systems, OSS sample applications, customer deployments, and LLM-connected developer tooling.
- Long-standing participant and contributor in Japan's engineering community since age 18, spanning technical blogging, hackathons, study groups, security programs, conference volunteering, alumni talks, and international technical competitions.
- Skilled at translating complex AI / edge-cloud technologies into working demos, OSS sample applications, technical write-ups, and practical engineering guidance.

SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Bash

AI / ML: PyTorch, TensorFlow, OpenCV, Computer Vision, Anomaly Detection, Edge AI

GenAI / LLM: Gemini API, Claude API, OpenAI API, LangChain, MCP framework, RAG, multimodal

Cloud & Infra: Azure, AWS, GCP, Docker, CI/CD, GitHub Actions, MLOps

Developer Advocacy: Technical writing, demo creation, public speaking, community building, English/Japanese technical communication

EXPERIENCE

Senior Software Engineer | Sony Semiconductor Solutions, Tokyo

Apr 2026 – Present

- Developing production-oriented AI camera software: image acquisition pipeline, C++ anomaly-detection inference, and PLC integration for factory environments; built full-stack operator web app
- Led evaluation and adoption of **generative AI tools, CI/CD pipelines, remote access infrastructure, and GitHub repository strategy** for a newly formed team

AI Solutions Engineer / Architect | Sony Semiconductor Solutions, Tokyo

Jan 2023 – Mar 2026

Edge AI Sample Applications & OSS

- Led **6-engineer team** to develop and OSS-release **9 Edge AI sample apps**; featured in exhibitions, national project demos, and enterprise briefings across Japan
- Drove international OSS collaboration (**Raspberry AI Camera**) with Lund (Sweden), Raspberry Pi Foundation, and Midokura (Spain)

Edge-Cloud AI Deployment

- Deployed multi-camera edge-cloud AI systems across **12 customer environments**; delivered AI solutions for **8+ enterprise clients** (license plate recognition, shelf analytics, digital signage)

Generative AI & Developer Tooling

- Led **5-engineer team** to build and OSS-release **AITRIOS MCP Server** — enabling LLM agents to interface with edge AI APIs; automated **~80% of API migration work** (V1 → V2) via LLM-assisted scripts

Edge AI Embedded Engineer | Sony Semiconductor Solutions, Tokyo

Jan 2020 – Dec 2022

- Developed SDK and first-gen Edge AI demos for Sony's **IMX500**; technical lead for **2 exhibition deployments** (system setup, live operation, on-site client briefings)
- Designed full-stack on-premise camera systems covering circuit review, SoC research, and HW validation

Computer Vision / ML Engineer | Sony Semiconductor Solutions, Tokyo

Apr 2019 – Dec 2019

- Independently built **46-class sign language recognition (98% accuracy, 15 FPS)** using ToF + facial landmark fusion — data collection to deployment
- Led ML dev of ToF+RGB 3D pose estimation; **exhibited at CES 2019**

EDUCATION

M.S. in Robotics

Tohoku University 2017–2019

MEMS sensors, semiconductor processes, data utilization, machine learning

B.Eng. in Mechanical Engineering

Tohoku University 2013–2017

Molecular robotics (DNA nanotechnology); published in *Small* (2017); robot contests, particle simulation (C++ / OpenGL)

COMMUNITY & ADVOCACY

Engineering Blogger — 55+ public blog posts and event reports covering AI, embedded systems, developer tools, and conference experiences

Accepted Speaker, Apache Community Over Code 2026 — Glasgow, UK; accepted talk at the Apache Software Foundation's international community conference

Gemini API Hackathon Mentor — Google AI Student Ambassadors program

SW Conference Staff — SRE NEXT, Open Source Summit Japan (NOC & venue ops, speaker support)

SecHack365 Alumnus Speaker — NICT 365-day security program (2019); invited as speaker for subsequent cohorts

ML Club Tohoku — Founding member, Whole Brain Architecture Young Researchers (2018)

10+ hackathons — NASA Space Apps Challenge, OpenCV AI Competition, humanoid robotics, edge AI, mobile apps and more

CERTIFICATIONS

Google Certified Professional – Cloud Architect

TensorFlow Developer Certificate

AWS Solutions Architect – Associate

Azure AI Engineer Associate

Registered Information Security Specialist (Japan)

AWARDS & PUBLICATIONS

SecHack365 (2019) — Built Fasttext AI to visualize emotional transitions; selected for SXSW Austin, won corporate prize at on-site hackathon

Grand Prize, BIOMOD 2015 — International biomolecular design competition, Harvard University

3rd Place Worldwide, iCAN'14 — International embedded systems competition (age 18)

Publications: Froemel, Akita et al. *Micromachines* (2020); Uchida, Akita et al. *Small* (2017)